A Posteriori Error Estimation

Residual functional and elementwise localization

Lecture outline

This lecture begins with the finite element solution already computed.

1. Define its residual and connect the residual to the unknown error.
2. Explain why the exact residual identity is not yet computable.
3. Introduce mesh facets, normals, and normal-flux jumps.
4. Begin localization by integrating by parts on each element.

This begins the *estimate* step of

$$\text{solve} \longrightarrow \text{estimate} \longrightarrow \text{mark} \longrightarrow \text{refine}.$$

The next lecture completes the residual representation and constructs the computable estimator.

Residual and error

What can be computed after the finite element solve?

From an a priori rate to an a posteriori estimate

Chapter 7 established an a priori estimate of the form

$$\|u - u_h\|_a \leq C_I \sqrt{k_1}\, h^{s-1}\, |u|_{H^s(\Omega)}\,.$$

It predicts how the error should decrease under uniform refinement, provided the exact solution has the assumed regularity. It does not directly determine

- the error in the solution already computed, because $|u|_{H^s(\Omega)}$ is unknown; or
- which elements contribute most to that error, because the estimate contains the global mesh size h.

An a posteriori estimate replaces the unknown higher derivatives of u by quantities computed from u_h and the prescribed data.

Recall the continuous and discrete diffusion problems

With $\boldsymbol{q} = k\nabla u$ and $q = \boldsymbol{q} \cdot \boldsymbol{n}$, define

$$a(w, v) = \int_\Omega k\nabla w \cdot \nabla v \, \mathrm{d}x, \qquad \ell(v) = \int_\Omega fv \, \mathrm{d}x + \int_{\Gamma_N} qv \, \mathrm{d}s.$$

The exact solution satisfies

$$a(u, v) = \ell(v) \qquad \forall v \in \mathcal{V}_0,$$

whereas the finite element solution satisfies

$$a(u_h, v_h) = \ell(v_h) \qquad \forall v_h \in \mathcal{V}_{h,0} \subset \mathcal{V}_0.$$

Both equations use the same forms. Their difference is the set of test functions: the discrete equations test only with functions in $\mathcal{V}_{h,0}$.

The residual measures what the discrete equations do not test

For any $v \in \mathcal{V}_0$, define

$$\boxed{R(u_h; v) = \ell(v) - a(u_h, v).}$$

This number is computable from u_h, k, f, and q for any selected test function v.

The exact equation gives $\ell(v) = a(u, v)$, so with $e = u - u_h$,

$$\begin{aligned} R(u_h; v) &= a(u, v) - a(u_h, v) \\ &= a(u - u_h, v) \\ &= \boxed{a(e, v)}. \end{aligned}$$

For $v_h \in \mathcal{V}_{h,0}$, the discrete equation gives

$$R(u_h; v_h) = 0.$$

Thus the residual vanishes on the discrete test space, but generally not on all of $\mathcal{V}_0$.

The residual dual norm equals the energy error

Because $a(\cdot,\cdot)$ is an inner product on $\mathcal{V}_0$,

$$|R(u_h; v)| = |a(e, v)| \leq \|e\|_a \, \|v\|_a .$$

Dividing by $\|v\|_a$ and taking the supremum gives

$$\sup_{0 \neq v \in \mathcal{V}_0} \frac{|R(u_h; v)|}{\|v\|_a} \leq \|e\|_a .$$

If $e \neq 0$, choose $v = e$. Then

$$\frac{|R(u_h; e)|}{\|e\|_a} = \frac{a(e, e)}{\|e\|_a} = \|e\|_a .$$

Therefore

$$\boxed{\|u - u_h\|_a = \sup_{0 \neq v \in \mathcal{V}_0} \frac{|R(u_h; v)|}{\|v\|_a}.}$$

Why the exact residual identity is not yet computable

The identity

$$\|e\|_a = \sup_{0 \neq v \in \mathcal{V}_0} \frac{|R(u_h; v)|}{\|v\|_a}$$

is exact, but it is not yet an estimator that can be evaluated:

1. the supremum ranges over the infinite-dimensional space $\mathcal{V}_0$;
2. equality is attained in the unknown direction $v = e$; and
3. the resulting single number does not identify where the error is produced.

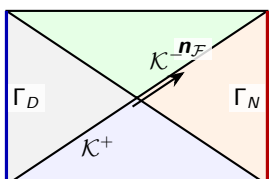
We therefore need two further steps:

localize the residual	and	bound its action on an arbitrary v.

Localizing the residual

Separate balance defects inside elements from flux defects on facets.

A conforming mesh has three facet classes



- $\mathcal{F}_h^{\mathrm{int}}$: interior facets shared by two elements;
- $\mathcal{F}_h^D$: boundary facets on Γ_D;
- $\mathcal{F}_h^N$: boundary facets on Γ_N.

On an interior facet, label the neighboring elements $\mathcal{K}^+$ and $\mathcal{K}^-$. Choose $\boldsymbol{n}_{\mathcal{F}}$ to point from $\mathcal{K}^+$ to $\mathcal{K}^-$. This fixes an orientation for the facet.

The next slide distinguishes a scalar jump from the complete normal-flux jump used in the residual.

The normal-flux jump includes the chosen normal

For a piecewise scalar quantity w, the oriented jump is

$$[\![w]\!]_{\mathcal{F}} = w^{+} - w^{-}.$$

Reversing the element labels changes its sign:

$$w^{-} - w^{+} = -[\![w]\!]_{\mathcal{F}}.$$

For a piecewise vector field $\boldsymbol{p}$, the quantity needed after integration by parts is the **normal-flux jump**

$$[\![\boldsymbol{p} \cdot \boldsymbol{n}_{\mathcal{F}}]\!]_{\mathcal{F}} := (\boldsymbol{p}^{+} - \boldsymbol{p}^{-}) \cdot \boldsymbol{n}_{\mathcal{F}}.$$

If the labels are reversed, the chosen normal must also be reversed. Therefore

$$(\boldsymbol{p}^{-} - \boldsymbol{p}^{+}) \cdot (-\boldsymbol{n}_{\mathcal{F}}) = (\boldsymbol{p}^{+} - \boldsymbol{p}^{-}) \cdot \boldsymbol{n}_{\mathcal{F}}.$$

The two sign changes cancel. Only this complete normal-flux expression is independent of the labeling.

First local step: integrate by parts on one element

The gradient term in the residual is a sum over elements:

$$\int_\Omega k\nabla u_h \cdot \nabla v \, \mathrm{d}x = \sum_e \int_{\mathcal{K}_e} k\nabla u_h \cdot \nabla v \, \mathrm{d}x.$$

On one element, Green's formula gives

$$\begin{aligned}\int_{\mathcal{K}_e} k\nabla u_h \cdot \nabla v \, \mathrm{d}x = & -\int_{\mathcal{K}_e} \nabla\cdot(k\nabla u_h) v \, \mathrm{d}x \\ & + \int_{\partial\mathcal{K}_e} (k\nabla u_h \cdot \boldsymbol{n}_e) v \, \mathrm{d}s,\end{aligned}$$

where $\boldsymbol{n}_e$ is the outward normal of $\mathcal{K}_e$.

This operation moves the derivative from v onto the computed flux. It creates an element-interior term and one boundary term for every element.